CAPSTONE · AI FOR GOOD · WEEK 5

The grid is the bottleneck, *not the turbines*

Europe can build wind farms faster than it can build the lines, cables and substations that carry their power. Tonight: why that is so, what it costs, what the EU is doing about it, and what it means for your four worlds.

Dr. Jan Erik Meidell · Wednesday 4 November 2026 · online · 18:15–20:00 · keep the shared canvas open on a second screen

TONIGHT

One hour forty-five, in five parts

18:18	What a grid is, who builds it, and why it is slower than a wind farm
18:30	The connection queue: how big, and why it forms
18:40	What congestion costs: redispatch, negative prices, the capture rate
18:55	Interconnection, and the Iberian blackout
19:02	Exercise 1 · where does each of your worlds hit the grid?
19:15	Break · 5 minutes
19:20	The EU's answer: the European Grids Package, and what it does not solve
19:34	The balancing loop, and the issues to 2030–2040
19:42	Exercise 2 · one grid indicator per world
19:53	Into Phase 1, due Sunday · close

VOTE IN THE CHAT · T OR W

What slows European wind more today: the turbines or the wires?

T: building enough turbines. W: building enough wires to connect them and carry their power.

Tonight's answer: **the wires**. The International Energy Agency counts at least **3,000 GW** of renewable projects waiting for a grid connection worldwide, 1,500 GW of them already advanced. That is about five times the wind and solar the world added in 2022.

IEA, *Electricity Grids and Secure Energy Transitions*, October 2023

WHAT A GRID IS

Three layers, two kinds of owner, one bill

Generation

Wind farms, solar parks, gas and nuclear plants. Built by **companies that compete**: Ørsted, Eni's Plenitude, hundreds of others. They decide where and when to build.

Transmission

High-voltage lines and the offshore export cables. Built by a **transmission system operator (TSO)**, one per country or region: TenneT, 50Hertz, Terna, Red Eléctrica. A regulated monopoly.

Distribution

The medium- and low-voltage network to homes and factories. Built by **distribution system operators (DSOs)**, about 2,500 in the EU, also regulated monopolies.

The EU grid is over **11 million kilometres** long. Generators choose where to build; the network companies must then connect them, and consumers pay for the network through their bills.

European Commission, *European grids*, DG Energy topic page, read 4 November 2026

WHY THE GRID LAGS THE FARMS · REASON 1 OF 3

A wind farm is quicker to build than the line it needs

Renewable project

1–5 years

to plan, permit and build

Transmission line

5–15 years

to plan, permit and build

A line crosses many municipalities and many landowners; each can object. A wind farm sits on one site. EU law now caps permits for renewables at 12 months onshore and 2 years offshore inside acceleration areas. No such cap makes a 700 km line quick.

IEA, 2023 · RED III, Directive 2023/2413

WHY THE GRID LAGS THE FARMS · REASON 2 OF 3

A regulated company may only build what it can prove is needed

How a TSO or DSO is paid

It does not sell power. The regulator sets a tariff that lets it recover the cost of lines it has built, plus a return. A line the regulator judges unnecessary is not paid for. So the operator waits for firm connection requests before it builds.

What that produces

The line is started when the farm is already planned, and finished years after the farm. The Commission calls the remedy **anticipatory investment**: building the line before the farms ask, on the basis of a plan, and letting the operator recover the cost. Guidance was issued on 2 June 2025.

European Commission, *Guidance on anticipatory investments for developing forward-looking electricity networks*, 2 June 2025; stakeholder workshop with ACER, 11 March 2026

WHY THE GRID LAGS THE FARMS · REASON 3 OF 3

Nobody wants the line through their own valley

SuedLink, Germany

Most German wind is in the north and in the North Sea. Much of the demand is in Bavaria and Baden-Württemberg. SuedLink, a 700 km direct-current line between them, was due in **2022**. It is now due at the end of **2028**, at about **€10 bn**, partly because it was moved underground after local protests.

What this means for a scenario

Public acceptance is a constraint with a number attached: six years and several billion euros on one line. The Commission's package names "public acceptance and benefit-sharing" as a condition of faster permitting, because a permit granted against the local population is appealed.

AN ØRSTED EXAMPLE

Borkum Riffgrund 3: the farm was built, the cable was not

913 MW, Germany's largest offshore wind farm, about 72 km off the coast: the turbines were up, but the grid operator's export link (TenneT's DolWin5) was not. Commercial operation slipped to early 2026.

Who builds what

In Germany and the Netherlands the transmission system operator builds the offshore connection; the developer builds the farm. Two companies, two timetables, one revenue stream.

What the developer loses

Every month of delay is a month of capital spent and no power sold. Week 4: the cost of capital is the biggest line in an offshore wind farm's costs.

Ørsted's film on the farm is on the Week 5 page. Delay as reported in the trade press, 2025: check the current status in Ørsted's latest interim report before you cite it.

THE SIZE OF THE JOB TO 2030

What the EU says its own grid needs this decade

€584 bn

of grid investment needed in the EU this decade, most of it in local distribution grids

+60%

electricity consumption in the EU is expected to rise by about 60% by 2030, as heating, transport and industry electrify

40%

of distribution grids are more than 40 years old

×2

cross-border transmission capacity is due to double by 2030

European Commission, EU Action Plan for Grids, COM(2023)757, 28 November 2023, as restated on the *European grids* topic page, 2026

THE QUEUE

Projects waiting for a connection, in numbers

WHERE	WHAT	SOURCE
EU	Connection queues exist in at least **16 EU countries** (mid-2026). About **120 GW** of mature renewable projects, including **1.5 million household installations**, risk not getting grid access in time for 2030.	Ember, 2026, cited by the Commission
Great Britain	Over **738 GW** in the queue before reform. "First come, first served" became **"first ready, first connected"** (approved April 2025): projects must show land rights and planning, and fit the 2030 plan. The reformed queue: **381.5 GW**, about half.	NESO

A queue is not a pipeline. Many projects in it will never be built, and ready ones wait behind speculative ones.

WHY QUEUES FORM

The Commission names four root causes

1. **Inadequate planning and slow network development.** The network is sized for yesterday's requests (reasons 1–3).
2. **No transparency on available capacity.** A developer cannot see where the grid has room, so it applies where the wind is, not where the line is.
3. **No locational incentive.** Connecting in a congested area costs the developer the same as connecting where there is spare capacity.
4. **Procedures that ignore project maturity.** A project with no land and no permit holds its place in the queue as long as one that is ready to build.

European Commission, *European grids* topic page: "multiple root causes… inadequate planning and delays in network development, lack of transparency on available capacity and locational incentives for users, and grid connection procedures not accounting for maturity"

WHAT THE EU TELLS MEMBER STATES TO DO ABOUT THE QUEUE

Four rules for the queue, and one new kind of contract

Guidance on grid connections, C/2025/8473

- **First ready, first served** instead of first come, first served
- **Transparent maturity criteria** for every request
- **Milestones with penalties**: a project that misses them loses its place
- **Regular cleaning** of the queue

Flexible connection agreements

A farm is connected now, on condition that the operator may reduce its output at the hours the line is full, instead of waiting years for a firm connection. Discussed at the first Commission workshop, 15 April 2026, together with hybrid sites (wind + solar + storage on one connection).

Guidance adopted with the European Grids Package, 10 December 2025. Great Britain's queue reform is the same principle.

WHAT CONGESTION IS

When the line is full, the operator pays one plant to stop and another to start

Redispatch

When the wind blows in the north and the north–south line is full, the German grid operator orders northern wind farms to reduce output and southern power plants to increase it. The switched-off farm is compensated; the southern plant is paid to run. Both payments go on consumers' bills.

Curtailment

The same thing seen from the wind farm: output it could have sold and did not. Whether it is paid for depends on the country and the contract. Uncompensated curtailment is lost revenue.

Congestion is not a shortage of wind or of demand. It is a shortage of line between the two.

WHAT CONGESTION COSTS

Germany and Great Britain pay about €5 billion a year to manage congestion

	COST	VOLUME	SOURCE
Germany · managing congestion (redispatch, curtailment, reserves)	**€2.95 bn (2024) · €3.08 bn (2025)**	**30.3 TWh a year**	Bundesnetzagentur, via the Bundestag
Great Britain · constraint costs, financial year 2024/25	**£1.7 bn (+64%)**	**13.5 TWh**	NESO

30 TWh is about the yearly electricity use of Ireland. It is paid for by consumers, through their bills.

NESO projects balancing costs of about £8 bn a year by 2030 unless network projects are brought forward.

BY HAND

What 10% curtailment costs one wind farm

STEP	CALCULATION	RESULT
A 1,000 MW offshore farm at a 45% capacity factor	1,000 MW × 8,760 h × 0.45	**3,942 GWh a year**
The grid can take only 90% of it	3,942 × 10%	**394 GWh lost**
At €60 per MWh	394,200 MWh × €60	**≈ €23.7 m a year**

Whether that €23.7 m is lost depends on the contract: compensated curtailment protects the owner, uncompensated curtailment does not. Read the contract before you read the forecast.

Try it with your own numbers in the calculator on the Week 5 page.

NEGATIVE PRICES

Why the price of electricity falls below zero, and how often

GERMANY	2023	2024	2025
Hours with a negative day-ahead price	301	457	573
Share of the year's 8,760 hours	3.4%	5.2%	6.5%

The mechanism: on a sunny, windy, low-demand hour there is more power than the grid can use, store or export. Plants that cannot switch off quickly, and producers paid a subsidy per MWh, keep running. The only way to balance the market is a price below zero: producers pay to deliver.

EPEX Spot (2023), SMARD / Bundesnetzagentur (2024–2025). Spain had about 555 negative hours in 2025, and 397 in the first quarter of 2026 alone.

THE CAPTURE RATE

Wind earns less than the average price, because all wind farms sell at the same hours

By hand · Germany, 2025

Average spot price: **8.932 ct/kWh** (€89.32/MWh).
What onshore wind actually earned on average: **7.441 ct/kWh**.
7.441 ÷ 8.932 = **83%**. That ratio is the **capture rate**.

Why

All wind farms produce at the same windy hours, so they push the price down exactly when they sell. The more wind on the system, the stronger the effect. Solar is hit harder: about **50%** in Germany in 2025.

A 2040 world with twice the wind and no new lines, storage or flexible demand is a world with a lower capture rate. That is a revenue forecast, not a weather forecast.

Market values: Rödl & Partner from the German market-value data, February 2026; the ratio is computed. S&P Global gives about 82% for 2024.

INTERCONNECTION · THE EU TARGET IS 15% BY 2030

How much of its surplus a country can send to its neighbours

MEMBER STATE	INTERCONNECTION, JAN 2026
Cyprus · Ireland	**0%**
Spain	**3.11%**
Poland	**5.06%**
Italy (Eni's home market)	**5.13%**
France	**5.44%**
Germany	**9.37%**

Cross-border capacity as a share of a country's generating capacity. 16 member states are above 15%; 9 are below 10%. A country below 10% has to curtail or accept negative prices at home when it has a surplus. European Commission, electricity interconnection targets.

28 APRIL 2025, 12:33

The Iberian blackout: what the investigators found, and what they did not

Spain and Portugal lost power almost entirely. The final expert report (ENTSO-E, March 2026) found no single cause: oscillations, gaps in voltage control and differing voltage practices led to fast voltage increases and a cascade of plants disconnecting.

What it does not say

It does not blame "too much solar". That claim is common in the press and is not what the investigators found; do not put it in your report.

Why it matters here

A grid with more wind and solar needs more **control**: voltage, inertia, coordination, links to neighbours. Spain is 3.11% interconnected. Resilience is one of the Grids Package's five aims.

SORTING

Turbine problem or grid problem?

1. A finished offshore farm waits a year for its export cable. **Grid**
2. Blades fail on a new turbine model and must be replaced. **Turbine**
3. Scottish farms are paid to stop because the line south is full. **Grid**
4. Wind earns 83% of the average price. **Grid: too little flexibility to move the surplus**
5. A project waits years in a connection queue. **Grid**

EXERCISE 1 · 13 MINUTES · BREAKOUT ROOMS

Where does each of your worlds hit the grid?

1. Take your team's **four worlds** from Week 4, with their 2040 wind numbers.
2. For each world, in the canvas table **"Where the grid binds"**: the **first** grid constraint it meets (connection queue, north–south lines, interconnection, negative prices, cable supply) and roughly **when**.
3. One sentence: does the world's 2040 number **survive** that constraint? If not, change the number or the world.

Back in the main room at 19:12. One person per team reads its most surprising row.

READ-OUT

Which world did the grid change?

A world that assumes 250 GW offshore by 2040 also assumes the lines, platforms and cables to land it. If your world does not say who builds them, who pays for them and when, it is a wish list.

This is report §4, the bottlenecks: where each one binds, and at what deployment level.

BREAK

Five minutes

Back at 19:20. Cameras on when you return.

THE EU'S ANSWER · 10 DECEMBER 2025

The European Grids Package: two laws and two guidance documents

What it is

COM/2025/1005, presented 10 December 2025, two years after the Action Plan for Grids. A proposal to **revise the TEN-E Regulation** (the law on cross-border energy networks) and a proposal to **speed up permits** by amending the Renewable Energy Directive, the Electricity Market Design and the Gas Directive. Consultation 13 May – 5 August 2025.

Its five aims, in the Commission's words

1. EU-level coordination to **map and plan** the grids needed
2. Tools for **cost-sharing**, so projects are funded "in a fairer and more equitable way"
3. **Faster permitting** for grids, renewables, storage and charging, "while ensuring public acceptance and benefit-sharing"
4. **Existing infrastructure used better**: new technology, flexibility, storage
5. **Resilience and security** of cross-border infrastructure

European Commission, *European grids*, DG Energy topic page, read 4 November 2026. These are proposals: Parliament and Council must adopt them before they bind anyone.

THE PACKAGE · THE EIGHT ENERGY HIGHWAYS

Eight bottlenecks the Commission wants fast-tracked

Electricity

- **Iberian Peninsula**: two crossings of the Pyrenees, so Spain's surplus can reach France
- **Great Sea Interconnector**: Cyprus linked to the continent (today 0%)
- **Harmony Link**: the Baltic states, independence from the Russian grid
- **Bornholm Energy Island**: a Baltic offshore hub, 3 GW, Denmark and Germany sharing the subsidy
- **South-East Europe**: price stability and storage

Gas and hydrogen

- **TransBalkan Pipeline**: reverse gas flow for the Balkans
- **SouthH2 Corridor**: Tunisia – Italy – Austria – Germany
- **Southwest hydrogen corridor**: Portugal to Germany

Fast-tracking means targeted support to the member states and promoters, help with financing, and streamlined permitting.

Four of the five electricity highways are on this course's map: Spain's export problem, Italy's corridor, Ørsted's Baltic market. Same source as the previous slide.

THE PACKAGE · WHO PAYS

Two tools for the money: cross-border cost-sharing and two-way contracts for difference

Cost-sharing between countries

An interconnector between Denmark and Germany is paid for by Danish and German consumers, but its benefits (cheaper power, less curtailment) fall unevenly. Until Bornholm, each state paid for its own side; Bornholm is the first project where two states **share one subsidy**. The revised TEN-E is meant to make that the rule, not the exception.

Two-way contracts for difference, C/2025/8479

A contract between a generator and usually a public body: when the market price is below the agreed price the state pays the difference, when it is above the generator pays it back. The generator's revenue no longer depends on the capture rate; the consumer no longer pays windfall prices. Guidance on how to design them was adopted with the package.

For your report: under a two-way CfD, the capture-rate risk moves from Ørsted to the state. Under a market-price contract it stays with Ørsted.

THE GRID'S OWN SUPPLY CHAIN

Cables and transformers take years to buy, and a handful of factories make them

COMPONENT	TIME TO PROCURE	PRICE, REAL TERMS, SINCE 2019
Cables	**2–3 years**	**almost ×2**
Large power transformers	**up to 4 years**	**+75%**
High-voltage direct-current (HVDC) cables	**over 5 years**	**—**

Procurement times have about doubled since 2021. A faster permit does not shorten a five-year order book. The Action Plan's seventh area is "strengthening grid supply chains" for this reason.

IEA, 25 February 2025. Permitting is still the main cause of delay.

OFFSHORE GRIDS

The sea needs its own grid, and it is late

TenneT 2 GW programme

14 standard 2 GW direct-current links in the Dutch and German North Sea, due 2031–32, running late; part of one contract was terminated in October 2025.

Danish North Sea island

Tender paused in June 2023: "too expensive and too risky".

Bornholm Energy Island

Paused in January 2025; revived in January 2026 as a 3 GW Danish–German project, with the two states sharing the subsidy. Now one of the eight Energy Highways.

Offshore grids are large, shared and political: whoever pays decides whether they are built. Ørsted's 2040 depends on these three programmes more than on its own turbines.

WEEK 2'S BALANCING LOOP, WITH TONIGHT'S NUMBERS

The loop that slows the transition

More wind → more congestion (+) → more curtailment and negative prices (+) → lower revenue per MWh (−) → less investment (+) → less wind (+).

One minus sign: an odd number, so it is a **balancing** loop. It does not stop growth; it slows it, and it gets stronger as wind grows. Tonight's numbers are the loop measured: 573 negative hours, an 83% capture rate, €3 bn of redispatch.

This is the second loop your report §5 needs, next to the reinforcing learning-curve loop. Write it in the A → B (+) notation.

WHAT WEAKENS THE LOOP, AND WHICH OF IT THE PACKAGE ADDRESSES

Five ways out, none of them quick

Build lines

Transmission and interconnectors: the real fix, 5–15 years. The package: TEN-E revision, Energy Highways, anticipatory investment.

Store

Batteries move power by hours; they cut negative-price hours, not north–south distance. The package: storage in the permitting fast lane.

Shift demand

Electric heating, vehicles and industry that use power when it is cheap. The package: "flexibility" under aim 4; the July 2026 bills proposal (COM/2026/600).

Faster permits

RED III deadlines for farms; the package extends fast permitting to lines and storage, with benefit-sharing for the people who live by them.

Price by location

Prices lower where power is in surplus, so new demand and plants move there. Debated, and rejected in Great Britain in 2025. The package asks for locational incentives in connection charges, not for zonal prices.

WHAT THE ISSUES WILL BE · 2030 TO 2040

Six problems your scenarios have to take a position on

1. **Who pays.** €584 bn this decade goes on bills as network tariffs. The Commission's July 2026 proposal on electricity bills exists because this is already visible to consumers. A world with fast grid build-out is a world with higher network charges first.
2. **Acceptance.** Lines through valleys and cables through fishing grounds are appealed. Benefit-sharing with local communities is now written into EU policy; whether it works is an open question.
3. **Cost allocation between countries.** Every interconnector and every offshore hub needs an agreement on who pays what. Bornholm took three years and one cancellation to reach one.
4. **Supply chain.** Cable and transformer order books of 2–5 years; a few factories; the same copper and steel the turbines need (Week 6).
5. **Security.** Cables on the seabed and a grid that runs on software; the package makes resilience one of its five aims.
6. **Flexibility.** If storage and flexible demand do not grow with wind, the capture rate falls and the balancing loop bites, whatever the lines.

YOUR TWO COMPANIES

Who carries the grid risk: Ørsted or Eni?

Ørsted · a generator

Sells power into the grid from large offshore farms. Exposed to connection delays (Borkum Riffgrund 3), to curtailment where it is not compensated, and to falling capture prices where it sells at market prices. A two-way contract for difference shifts much of that risk to the state.

Eni · Plenitude

A smaller renewable fleet, spread across technologies and countries, and a retail business that sells power to customers. In Italy it works in a grid with 5.13% interconnection. The question for your report: does owning the customer reduce the grid risk, or only move it?

The grid is a scenario variable. In a strong-policy world it is built faster and the loop is weak; in a weak-policy world farms are built where the grid cannot take them, and the loop bites. Grid build-out can be one of your two axes, if it is not just an outcome of your policy axis.

EXERCISE 2 · 9 MINUTES · BREAKOUT ROOMS

One grid indicator per world

1. Pick **one grid indicator** that would tell you which of your worlds is unfolding: negative-price hours (SMARD), constraint costs (NESO), interconnection % (European Commission), connection-queue GW (NESO, Ember), or an Energy Highway's completion date.
2. Write its **baseline** with year and source, and a **value in each world** for 2030.
3. Write it in the canvas box **"One grid indicator"**. It is one of the three per world that report §7 needs.

Back at 19:51. One indicator per team, read aloud.

READ-OUT

Does your indicator separate the worlds?

If all four worlds give it the same 2030 value, it cannot tell you which world you are in, however good the source.

PHASE 1 · DUE SUNDAY 8 NOVEMBER, 23:59

Where tonight goes in the report

TONIGHT	REPORT SECTION
Where each world hits the grid; the three reasons the grid lags; the queue and its causes	**§4 · The bottlenecks**
The balancing loop with its signs; who pays, who accepts, who shares the cost	**§5 · Stakeholders and the system**
Your grid indicator, with baseline and four values	**§7 · Indicators and datasets**

Sections 1–7, one voice, handed in from the report page by one of you. The Commission's *European grids* page is linked on the Week 5 page: what the EU says it will measure is a list of what you can measure.

CLOSE

Every 2040 wind number is also a statement about lines, cables, permits and who pays for them

Make yours say who builds the grid, who pays for it, and when. A number without those three is a wish, not a scenario.

Questions on the material: the forum of the week. Anything personal: the message panel, bottom right of every page.